# Deep Learning-Based Cloud Security: Innovative Attack Detection and Privacy Focused Key Management

Shahnawaz Ahmad, Mohd Arif, Shabana Mehfuz, Javed Ahmad, and Mohd Nazim

***Abstract*—Cloud Computing (CC) is widely adopted in sectors like education, healthcare, and banking due to its scalability and cost-effectiveness. However, its internet-based nature exposes it to cyber threats, necessitating advanced security frameworks. Traditional models suffer from high false positives and limited adaptability. To address these challenges, VECGLSTM, an attack detection model integrating Variable Long Short-Term Memory (VLSTM), capsule networks, and the Enhanced Gannet Optimization Algorithm (EGOA), is introduced. This hybrid approach enhances accuracy, reduces false positives, and dynamically adapts to evolving threats. EGOA is employed for its superior optimization capability, ensuring faster convergence and resilience. Additionally, Chaotic Cryptographic Pelican Tunicate Swarm Optimization (CCPTSO) is proposed for privacy-preserving key management. This model combines chaotic cryptographic techniques with the Pelican Tunicate Swarm Optimization Algorithm (PTSOA), leveraging the pelican algorithm's exploration strength and the tunicate swarm's exploitation ability for optimal encryption security. Performance evaluation demonstrates 99.675% accuracy, 99.5175% recall, 99.7075% precision, and 99.615% F1-score, along with reduced training (1.79s), encryption (0.986s), and decryption (1.029s) times. This research significantly enhances CC security by providing a scalable, adaptive framework that effectively counters evolving cyber threats while ensuring efficient key management.**

***Index Terms*—Cloud computing, attack detection, key management, pair key selection, deep learning, hybrid optimization.**

## I. Introduction

In the era of CC, businesses and individuals have embraced the promise of unparalleled scalability, cost-efficiency, and accessibility. This digital transformation, while offering tremendous advantages, has simultaneously expanded the attack landscape, exposing cloud environments to a multitude of evolving threats and security challenges. As organizations entrust their data, applications, and services to cloud infrastructure, safeguarding against a spectrum of malicious actors and potential vulnerabilities has become imperative in the relentless pursuit of secure and uninterrupted cloud operations [1]. Cloud infrastructures are appealing targets for potential attackers looking to exploit weaknesses and gain unauthorized access to valuable data. The implications of such cracks are significant, often resulting in data compromise, service disruption, and legal consequences [2].

Detecting and answering to attacks in real-time is imperative for preserving cloud systems. Security and privacy are at the forefront of concerns when operating in cloud environments. The confidentiality of data is of supreme importance, and cryptographic keys play a pivotal role in safeguarding these aspects. The compromises of cryptographic keys have catastrophic consequences, rendering encryption measures ineffective and exposing sensitive information to unauthorized access. Additionally, the secure generation and distribution of cryptographic solutions continue to be complex tasks, as they require robust protection against potential vulnerabilities. Thus, ensuring the secure generation, storage, distribution, rotation, and retirement of cryptographic keys is a critical challenge in cloud privacy. Privacy preservation poses difficulties when performing Attack Detection (AD), as it necessitates a delicate balance between security and user privacy [3], [4].

Conventional Machine Learning (ML) and Deep Learning (DL) approaches in AD within CC encompass a range of techniques. DL for Intrusion Detection Systems (IDS) leverages Convolutional Neural Networks (CNNs) [5] and Recurrent Neural Networks (RNNs) [6] to scrutinize network traffic and system logs for anomalies. While highly effective, they demand substantial labeled datasets and substantial computational resources, and they are susceptible to adversarial attacks. Generative Adversarial Networks (GANs) [7] are employed to generate attack scenarios and bolster the resilience of AD models, but they require expertise in GAN model tuning and are susceptible to training instability. Reinforcement learning [8] introduces adaptability into AD models, learning from the dynamic nature of cloud environments. Yet, challenges lie in reward design and model convergence. Long Short-Term Memory (LSTM) [9] networks excel in detecting attacks that unfold over time through their ability to model and analyze sequential data, though this approach demands intricate network architectures and

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significant historical data for training. Similarly, homomorphic encryption [10] enables computations on encrypted data without decryption, bolstering security. Nevertheless, its drawback is a high computational overhead, rendering it less practical for real-time processing.

Attribute-based encryption (ABE) [11] facilitates fine-access control by encrypting data with user attribute-based access policies, though it increases key management complexity. Threshold cryptography [12] spreads out key creation among several entities boosting security but requiring safe teamwork between key-sharing groups. All these current methods for secure and Privacy-Preserving Key Management (PPKM) in the cloud have made big steps forward but also face high computer workload issues, which cut down their speed and power. Some fixes can't keep up as cloud setups keep getting bigger and more complex. Also, privacy concerns don't always get the attention needed leaving room to do better at balancing security and protecting user info. New methods, like improved DL-based approaches, are becoming more popular as they show promise to boost the ability to spot and fight off security risks.

This research work aims to address these challenges by exploring innovative solutions that harness the power of optimized DL techniques and cryptographic frameworks. It seeks to overcome the confines of existing approaches, ultimately strengthening the security of cloud-based models in a rapidly evolving digital landscape. Below is a list of the key contributions.

**Contributions:**

- This research introduces a new Attack Detection (AD) model called VECGLSTM, which utilizes a hybrid approach incorporating a variable Long Short-Term Memory (VLSTM) network alongside a capsule network and an enhanced gannet optimization algorithm (EGOA).
- EGOA was selected over other algorithms due to its inherent capacity to efficiently handle complex optimization problems while providing a solid foundation for further improvements. This enhancement enables the algorithm to dynamically adjust its parameters during optimization, thereby enhancing its convergence speed and resilience in addressing various optimization challenges.
- Additionally, the research presents a Privacy-Preserving Key Management (PPKM) model named CCPTSO, which utilizes chaotic cryptographic techniques and a hybridized pelican tunicate swarm optimization algorithm. This contribution is noteworthy as it emphasizes the importance of protecting user privacy while implementing robust security measures, responding to the increasing concerns surrounding data privacy in the digital era.
- The combination of the Pelican Optimization (PO) algorithm's strong exploration capabilities and the Tunicate Swarm (TS) algorithm's robust exploitation abilities synergistically enhance the overall optimization performance. This approach attains an improved balance between exploration and exploitation, resulting in improved optimization outcomes.

**Organization:**

Section II reviews conventional approaches, detailing existing methodologies and their limitations. Section III introduces the new topology, describing its structure, features, and advantages over traditional methods. Section IV presents experimental outcomes and provides a comprehensive examination of the findings. Section V summarizes the key points, highlights the study's significance, and suggests future research directions.

## II. Review

This section assesses some of the recent research works among the various existing approaches in the cloud environment.

### A. Attack Detection Models

Protection Against Network Attacks using Cyber-security Ensemble Algorithms (PANACEA) was developed by Al-Essa et al. [13], utilizing a neural ensemble model to enhance cyber-threat detection. This approach improved IDS performance by leveraging machine learning techniques to increase accuracy and reduce false positives. A neural attention multi-output model, Robust Optimized Unified Learning Ensemble for Threat Evaluation (ROULETE), was proposed by Giuseppina Andresini et al. [14] to introduce explainability in network IDS. The model contributed to improving interpretability while maintaining high accuracy in threat evaluation. An ensemble method for Deep Neural Networks (DNNs) was developed by Folino et al. [15] to boost the efficiency of IDS. This approach integrated multiple neural networks, improving detection accuracy and robustness through ensemble learning techniques. A bagging-based gradient boosting decision tree model, Dual IDS (Dual-IDS), was introduced by Maya Hilda Lestari Louk and Bayu Adhi Tama, [16] for detecting network anomalies. The model optimized the detection of various attack types while maintaining computational efficiency. A Hybrid Intrusion Detection Model (HIDM) was designed by Vashishtha et al. [17] specifically for cloud-based systems, effectively identifying different types of attacks through the integration of signature and anomaly-based detection approaches. However, poor classification performance was observed on the UNSW-NB15 and CICIDS2017 datasets, indicating the need for further improvements. An attempt to mitigate attacks was made by Gulia et al. [18], who integrated ML techniques with the Artificial Bee Colony (ABC) approach, resulting in the Group-ABC (G-ABC) model. While this approach contributed to optimizing attack detection, only a limited number of attacks were detected. The Ensemble Intrusion Detection Model for Cloud Computing using Deep Learning (EICDL) was introduced by Salvakkam et al. [19] to improve IDS accuracy. Although IDS efficiency was enhanced, improvements in detection rate and accuracy were still required for practical implementation. To enhance IDS performance in cloud environments, Alzughaibi and El Khediri, [20] utilized a Multi-Layer Perceptron (MLP) with backpropagation and an MLP trained using Particle Swarm Optimization (MLP-PSO). However, real-time implementation proved to be challenging due to the active nature of cloud environments. A Hybrid Ensemble Supervised Learning (HESL) approach was developed by Samunnisa et al. [21], combining clustering and classification models for IDS implementation. This method enabled effective categorization of malicious attack types but resulted in

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significant increases in both running and training time, impacting scalability. The Hybrid Ensemble ML (HEML) model was designed by Saini et al. [22] for detecting various attack types using four datasets. While higher accuracy and efficiency were achieved, the selection of outdated or inappropriate datasets limited its effectiveness in real-time environments.

### B. Attack Detection With Key Management Models

The Whale with Cuckoo Search Optimization (WCSO) system was introduced by Kadry et al. [23] to improve attack detection accuracy and data protection. The system utilized a Quantum Neural Network (QNN) and Elliptical Curve Cryptography (ECC), significantly enhancing security measures. A Hybrid Enhanced Elman Spike Coot CNN (Hyb EESCCNN) was developed by Ahmad et al. [24] for attack detection within cloud environments. Additionally, the Adaptive Tangent Brakerski-Gentry Vaikuntanathan Homomorphic Encryption (ATBGVHE) method was proposed to strengthen system security. However, challenges were encountered in handling and distributing data within big data environments. A hybrid ECC-AES model was designed by Jammula et al. [25] to improve secure key generation and transfer. This approach demonstrated faster encryption compared to AES and other conventional cryptographic systems, contributing to enhanced security in IDS. An explainable deep learning-enabled IDS was developed by Marwa Keshk et al. [26] for IoT networks, emphasizing transparency and interpretability. The framework provided deeper insights into IDS decisions while maintaining high accuracy.

Arango-López et al. [27] have been setting models in place such that it has the flexibility and scalability of service-oriented, cloud-based architectures and hence, improve detection accuracy, as well as address the reconfiguration-related issues of modern infrastructures. However various solutions were presented to the enterprise in such a manner that they enjoy strong, easy, deployable security tools with a great deal of adaptation for different applications. The forward defense mechanisms have been added to strengthen the resilience of the network to the evolving cyber threat.

### C. Problem Statement

A pressing concern is the imperative need to assess the vulnerability of the cyber-attack surface and identify potential threat vectors. This emphasis is vital for the development of robust strategies aimed at mitigating cyber threats, bolstering resilience, and ensuring swift recovery in the event of an incident. A recent survey conducted by the 'Deloitte Center for Controllership' has unveiled a concerning landscape. It reveals that a significant 34.5% of executives who participated in the survey stated that their administrations' accounting and financial information fell prey to cyber adversaries over the past year. Among this cohort, 22% encountered at least one such cyber event, while 12.5% suffered multiple incidents. Equally concerning is the fact that nearly semi (48.8%) of C-suite and additional directors anticipate an escalation in both the frequency and magnitude of cyber events aimed at their organizations' economic information in the forthcoming years.

TABLE I
PROBLEMS RELATED TO ATTACKS AND CLOUD SECURITY

| **Challenges on attacks** | |
|---|---|
| **Problems related to attacks and cloud security** | **Percentage** |
| ◆ Experienced at least one cyber event | 22% |
| ◆ Suffered multiple cyber events | 12.5% |
| ◆ Anticipate escalation in cyber events | 48.8% |
| **Challenges to cloud security** | |
| ◆ Protecting data in cloud environments | |
| ◆ Managing key pairs in cloud environments | |
| ◆ Ensuring compliance and data residency | |

This challenge is compounded by the fact that only a scant 20.3% of respondents confirmed that their accounting and finance teams consistently collaborate closely with their cybersecurity counterparts. This confluence of factors depicted in Table I underscores the critical need for a comprehensive and proactive approach to cybersecurity and threat management to safeguard sensitive financial data in the modern digital landscape [28].

In essence, this research work aligns with the need for a comprehensive and proactive approach to cybersecurity and threat management mentioned in the problem statement. It strives to provide innovative solutions (for AD and PPKM in the cloud) to the challenges highlighted in the statement, ultimately contributing to the protection of data.

## III. INTRODUCED TOPOLOGY

This section explains the introduced topology encompassing AD and PPKM processes within cloud storage. Furthermore, the model incorporates the PPKM module to encrypt and decrypt normal data, enhancing data security.

### A. Data Acquisition

The input data for the introduced work is collected from four datasets namely, NSL-KDD [29], UNSW-NB15 [30], KDD-CUP [31], and CSE-CIC-IDS2018 [32]. All these datasets consist of several features with numerous types of network attacks. All these data are given for preprocessing. Class balance within the NSL-KDD and UNSW-NB15 datasets is achieved through a tailored preprocessing approach. The DRFLLS method is applied to manage missing values and maintain data consistency by estimating and selecting relevant features using CFS. This strategy effectively improved classification performance by reducing dimensionality. Furthermore, cluster assignments are optimized using the AHFLB clustering model, ensuring organized data and balanced representation across different classes.

The block diagram illustrated in Fig. 1 outlines stages including preprocessing, feature selection, and various AD techniques. The cloud environment in Fig. 1 is integrated to enhance secure key management in the proposed attack detection framework. It ensures a reliable and scalable infrastructure for handling cryptographic operations, such as optimal pair key selection, using the CCPTSO algorithm. By leveraging cloud capabilities, the system benefits from improved computational efficiency, secure storage, and real-time accessibility, facilitating seamless

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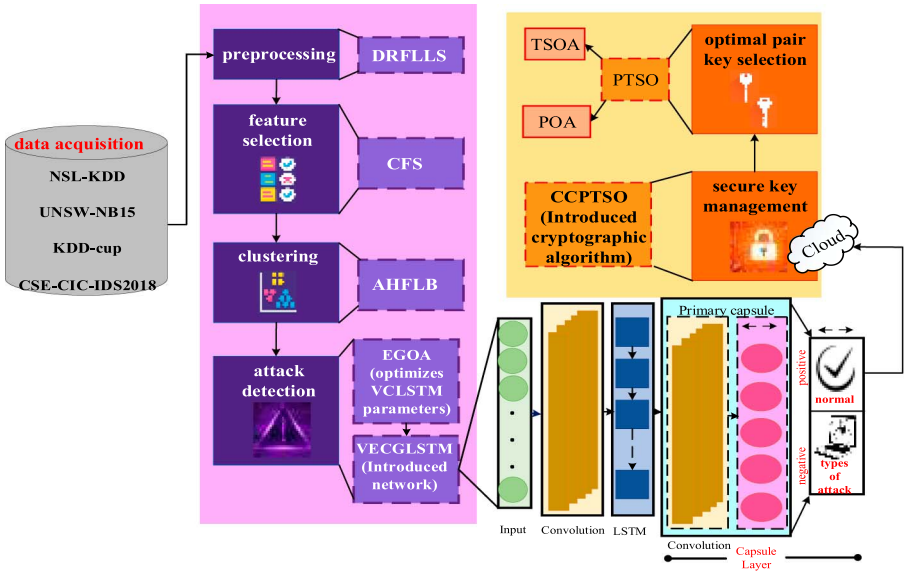



Fig. 1. Block diagram of introduced topology.

encryption and decryption processes. This inclusion strengthens data security, ensuring that cryptographic keys remain protected against potential cyber threats while maintaining efficient attack detection.

### B. Preprocessing of Data

Missing values further exacerbate the class imbalance issue by reducing the amount of available data for minority classes. Input data has missing and ineffective features, leading to erroneous conclusions and complicating user understanding. The Developed Random Forest with Local Least Squares (DRFLLS) method [33] is used to estimate missing values, which involves three steps:

Generate $r$ Similarity records with seven measures. Approximate missing values using LLS with the $r$ Values. Select the best estimate from seven options using Pearson correlation and Normalized Root Mean Squared Error (NRMSE). From the preprocessed output, the important and correct features are needed to be selected.

### C. Correlation-Based Feature Selection

A feature selection process based on correlation is executed following the normalization of the dataset. Correlation-based Feature Selection (CFS) [34] is a commonly used technique to choose the most valuable features. The feature selection using CFS is based on the following (1).

$$corr_{cs} = \frac{s \cdot corr_{ci}}{\sqrt{s + s(s-1)corr_{ii}}} \tag{1}$$

In this context, $corr_{cs}$, $corr_{ci}$ and $corr_{ii}$ represents the correlation, average correlation, and inter-correlation average between the class variable and the sum of the attributes within the subset features $s$. Commencing with an initial threshold of 0.5, any attribute exhibiting a correlation greater than 0.5 is selected for classification of attacks via clustering.

### D. Clustering of Selected Features

In cloud environments with multiple servers and resources, even with optimal feature selection, the Adaptive Hesitant Fuzzy Linguistic Bi-objective (AHFLB) clustering model [35] offers flexible cluster assignments using hesitant fuzzy linguistic information. It simultaneously optimizes two objectives: reducing intra-cluster distance and maximizing inter-cluster distance as stated in (2).

$$\begin{cases} min\left(\sigma = \sum_{c=1}^{C}\sum_{c_g=1}^{c_k} D(b^{c_g}, b^{c_c})\right) \\ min\left(s = \sum_{c=1}^{C}\sum_{c_g=1}^{c_k} |s^{c_g} - \overline{s_k}|\right) \end{cases} \tag{2}$$

This function considers the global optima of all clusters $C$, as opposed to the local optima of a specific cluster $c$, to ensure that the sum of distances $D$ among all clusters is minimized. $c_g$ is the specific point index within a cluster. $c_c$ is the total number of points in a cluster. $D(b^{c_g}, b^{c_c})$ is the distance between points $b^{c_g}$ and $b^{c_c}$. $s^{c_g}$ is the distance from a point $c_g$ to the centroid of another cluster. $\overline{s_k}$ is the mean distance between points in the cluster $c$ and centroids of other clusters. The clustering results are obtained by simultaneously minimizing $\sigma$ and $s$.

The calculation procedures of the clustering model are employed in the following steps.

- ➢ First, assess the steadiness of each preference relation using the multiplicative steadiness measure model. If a preference passes the test, proceed to the subsequent step. If a preference fails the test, remove it due to inconsistency and proceed to test the further step.
- ➢ Next, gather all the consistent preference relations by employing the AHFLB clustering approach, which simultaneously takes into account group consensus degree and

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group information entropy. Afterward, proceed to the following step.

➢ Merge preference information from various clusters by (3).

$$\acute{B}_{ij(l)} = I^{-1}\left(\sum_{c=1}^{C}\sum_{c_g=1}^{c_c}\frac{w_c}{n_c}\times I(b_{ij(l)}^{c_g})\right) \tag{3}$$

where $i,j = 1,2,...,n$ and $l = 1,2,3$, $\acute{B}_{ij(l)}$ is the value being computed for indices $i$, $j$, and $l$. $w_c$ is the weight associated with cluster $c$, $n_c$ Number of points in the cluster $c$. $I$ is the is the $I^{-1}$ is the inverse of the function $I$.

After clustering data from various sources within a cloud environment, the process of attack-type classification involves examining the clustered data to detect patterns or behaviors that deviate from normal operations. These anomalies are then classified into specific attack types using the introduced VCGLSTM network.

### E. Detection Module

This research work combines VLSTM with a capsule network [36] (VCLSTM) to improve training performance which creates a DL framework for detecting various types of attacks. The overall detection process is enhanced by optimizing the introduced network parameters with the EGOA. VECGLSTM leverages a sophisticated deep learning architecture that includes capsules and an enhanced long short-term memory network, which allows for better handling of spatial and temporal data. This enables the approach to accurately capture intricate patterns in the data, leading to more effective detection of both identified and unidentified cyber threats. The inclusion of the Gannet mechanism further refines the model's capability to adapt to new and evolving threats, ensuring high accuracy and robustness in threat detection.

❖ *VCLSTM:* To construct this structure, the cell and hidden state of the LSTM are initially represented as a group of capsules. This means that the values. $H_t$, $C_t$, which exist in the space $\mathcal{R}^{d_h}$, transform into $h_t$, $c_t$, residing in a space $\mathcal{R}^{n_c \cdot d_c}$. In this context, $n_c$ and $d_c$ are represented as the overall capsules and the dimensions of each capsule.

$$f_t^{j/i} = \sigma\left(w_f^{j/i}X_t + u_F^{j/i}h_{t-1}^i + \ B_F^{j/i}\right) \tag{4}$$

$$i_t^j = \sigma\left(w_i^j X_t + \sum_i u_I^{j/i}h_{t-1}^i\right) \tag{5}$$

$$o_t^j = \sigma\left(w_o^j X_t + \sum_i uh_{t-1}^i\right) \tag{6}$$

$$\tilde{c}_t^j = tanh\left(w_c^j X_t + \sum_i u_C^{j/i}h_{t-1}^i\right) \tag{7}$$

$$c_t^{j/i} = i_t^j \ \odot \ \tilde{c}_t^j + f_t^{j/i} \odot c_{t-1}^i \tag{8}$$

$$c_t^j = Routing\left(\left\{c_t^{j/i}\right\}_i\right) \tag{9}$$

$$h_t^j = o_t^j \odot c_t^j \tag{10}$$

where $f_t^{j/i}$, $i_t^j$, $o_t^j$, $h_t^j$ and $c_t^j$ in (4), (5), (6), (7), (8), (9) and (10) represents the forget, input, output, hidden gates and cell state. $w_f^{j/i}$, $w_i^j$, $w_o^j$ and $w_c^j$ represents the weight matrices of forget, input, output gates, and cell state. $B_F^{j/i}$ is the forget gate bias function.

In the whole training process, the margin loss function is adopted as the loss function $l_j$ as in (11).

$$l_j = t_j maximum\left(0, fp - v_j\right)^2 + \mu\left(1 - t_j\right)maximum\left(0, v_j - fn\right)^2 \tag{11}$$

where $j$ represents the classified outputs (if $j$ exists, its class is 1, and if $j$ not exists, its class is 0). $v_j$ and $t_j$ is the probability of occurrence and classification indicator function. $fp$ and $fn$ is the false positives and false negatives. The proportional coefficient is denoted by $\mu$.

In order to boost the classification accuracy, the forget gate bias of (4), weight matrices of (4) to (7), and the loss function of (11) are tuned by the EGOA.

❖ *Tuning Network Parameter:* EGOA is the enhanced version of the Gannet Optimization Algorithm (GOA) [37] allows for the exploration of the best results for a specific population. Specifically, EGOA introduces significant advancements in tuning parameters like forget gate bias ($B_F^{j/i}$), weight matrices ($w_f^{j/i}$, $w_i^j$, $w_o^j$ and $w_c^j$) and the loss function ($l_j$).

EGOA operates by dividing the population into five distinct sets, each conducting computations independently. This division allows for the simultaneous exploration of different regions within the solution space, enabling efficient management of diverse problem aspects in parallel. By leveraging this parallel processing approach, EGOA substantially reduces computation time, particularly beneficial for handling complex tasks in cloud security and attack detection. Moreover, EGOA enhances population diversity by maintaining separate Perturbation Vectors (PVs) for each set. These PVs are updated iteratively based on performance within their respective sets, distinguishing winners and losers to drive evolutionary progress. This strategy prevents premature convergence to suboptimal solutions, thereby increasing the likelihood of discovering high-quality solutions across the entire search space. This capability is critical in optimizing the parameters of VECGLSTM, ensuring robust performance in detecting evolving cyber threats with superior accuracy and efficiency.

EGOA's innovative approach to parameter tuning through parallel computation and adaptive population management enhances the optimization process, promoting faster convergence to optimal solutions.

To distinguish the winners and losers (considered as attack and non-attack data) in each set, it is denoted as the winner $i$ $(i = 1, 2, 3, 4, 5)$ and loser $(i = 1, 2, 3, 4, 5)$. The exact procedure is explained below:

a) The total population is divided into five sets, and $pv_i$ is initialized for every individual set, with $\sigma i = 10$ and $\mu i = 0$ where, $i = 1, 2, 3, 4, 5$.
b) Solutions $x_i$ are generated using $pv_i$, and corresponding solutions $x$ are generated using the PV of each set.

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c) Comparison between $x$ and $x_{new}$ is made for each set, and the winners and losers of each set are determined using the "compete" function $\{(winner,\ loser) = compete\ (x, x_{new})\}$.
d) Determine the objective function by (12) tuning the weight parameter and minimizing the loss and bias functions.

$$OF = tuning\left(w_f^{j/i}, w_i^j,\ w_o^j, w_c^j\right),\ \text{minimizing}\left(l_j, B_F^{j/i}\right) \quad (12)$$

e) The position update equation of the GOA is applied to $x$ in each set to generate a new position $x_{new}$.
f) PV is updated, and the best solution for every set and the universal finest solution is updated according to (13) and (14).

$$mx_i(IT+1) = \begin{cases} X_i(IT) + T * \odot * (X_i(IT) - X_B(IT)), & capturability \geq k(x) \\ X_B(IT) - (X_i(IT) - X_B(IT)) * T * levy, & capturability < k(y) \end{cases} \quad (13)$$

$$\Delta = capturability * |X_i(IT) - X_B(IT)| \quad (14)$$

where $X_i(IT)$ is the initial solution, $X_B(IT)$ is the optimal solution, *levy* signifies the levy flight of the gannet and $k$ represents a constant value of 0.2.
g) If the termination criteria ar satisfied, the algorithm is terminated; else, repeat Steps 2 through 5.

Hence, EGOA optimizes network parameters for accurate attack detection, followed by secure encryption and decryption in the key management process.

### F. Chaotic Cryptography-Based Privacy Preservation Model

A privacy preservation model based on cryptography is introduced to enhance the security of computing devices. The encryption process employs the introduced CCPTSO to select optimal key pairs and generate cipher text. Subsequently, decryption is performed on the cipher text using a similar best key to recover the unique content.

CCPTSO enhances the security and privacy aspects of the system by integrating chaotic cryptographic techniques with the PO and TSO. The chaotic cryptographic approach introduces a higher level of unpredictability and complexity in the encryption process, making it more resistant to attacks. CCPTSO optimizes key management, ensuring efficient and secure encryption and decryption processes. Together, these components provide a comprehensive solution that not only boosts the accuracy and robustness of threat detection but also significantly enhances data privacy, addressing the limitations of existing approaches and making the proposed technique highly effective in cloud security applications.

❖ *Data Collection:* The initial stage of this procedure involves the gathering of a necessary set of unaltered data (text data). These unaltered messages contain essential information earmarked for privacy preservation.

❖ *Formation of the Plain Text:* Proceeding to the following stage, plain text is generated from the accumulated information. To form the plain text, every character in the data is converted into a 16-digit binary value. Consider the tex $t$ with a character length $C_l$ intended for secure preservation. Upon adapting message $t$ into plain text $P$, every character is changed into a 16-digit binary value, resulting in plain text $P$ having a length $P_l$ determined $P_l = C_l \times 16$.

❖ *Chaotic Map Based Key Generation:* Following the creation of plain text, the generation of an optimal key based on a hybrid chaotic map is conducted to enhance the privacy preservation procedure. In this context, the hybrid chaotic map incorporates the features of dual distinct maps [38]: the circle map and the henon map. This chaotic map is formed through a nonlinear integration of the circle and henon maps, resulting in a 2D map. It is signified by (15) and (16) as follows.

$$U(i+1) = C(\gamma, \eta, v(i)) mod 1 \quad (15)$$

$$v(i+1) = H(\gamma, U(i)) \quad (16)$$

where $C(\gamma, \eta, v(i))$ refers to the Circle map, and $H(\gamma,\ U(i))$ represents the henon map, correspondingly. The variables $\gamma$ and $\eta$ are pivotal in controlling the chaotic behavior within the key.

Both the henon and the circle map display characteristics of chaotic behavior, rendering them fitting for merging to improve the chaotic map. Following the application of the circle and henon maps to the chaotic map, a 2D hybrid chaotic map is derived, with its representation found in (17) and (18), respectively.

$$U(i+1) = U(i) + \eta + (\gamma sin(2\pi V(i))) mod\ 1 \quad (17)$$

$$V(i+1) = 1 - \gamma U(i)^2 + V(i) \quad (18)$$

The (17) and (18) provided above illustrate the 2D hybrid chaotic map, with its chaotic behavior contingent on the variables $\gamma$ and $\eta$. In this research work, both parameters are adjusted using the introduced CCPTSO to ensure that key generation through the hybrid chaotic map achieves the highest level of confusion and diffusion.

❖ *Optimal Key Pair Selection:* In this section, the variables $\gamma$ and $\eta$ of the hybrid chaotic map are subjected to an optimization procedure by CCPTSO aimed at maximizing confusion and diffusion.

*Pelican Tunicate Swarm Optimization:* In the PO algorithm's initial exploration, the position doesn't match the introduced concept, leading to suboptimal solutions. The TSO faces performance issues due to collisions, especially with large or real-time data, affecting its results. To address these challenges, a new position update mechanism, the hybrid CCPTSO is introduced by hybridizing PO [39] and TS algorithm [40].

The position updation of the PO algorithm and TS algorithm are represented as $x_i$ and $\vec{t}_p$.

Initially, the difference between two positions is calculated using (19).

$$Var = x_i + \vec{t}_p \quad (19)$$

$$F = \frac{Variance}{2} \quad (20)$$

where $x_i$ and $\vec{t}_p$ represents the updated position of the PO and TSO algorithm. The difference between these two values computes the final position updation $F$ in (20). The optimal value is thus obtained by (20) which selects the optimal final best key pairs. Finally, encryption is performed using the best key pair, resulting in the creation of cipher text. The

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TABLE II
EXPERIMENTAL SETUP

| S.No | Parameters | Range |
|---|---|---|
| 1 | Filters count | 64, 128, 256, 512 |
| 2 | Kernel size | 3*3 |
| 3 | Dropout size | 0.5 |
| 4 | Learning size | 0.0001 |
| 5 | Iterations count | 100 |
| 6 | Optimizer for attack detection | EGOA |
| 7 | Optimizer for optimal key selection | CCPTSO |

TABLE III
CONFUSION MATRICES OF VARIOUS DATASETS (A) NSL-KDD, (B) UNSW-NB15, (C) KDD-CUP, AND (D) CSE-CIC-IDS2018

| Datasets | True Label\Predicted Label | Normal | Attacks |
|---|---|---|---|
| NSL-KDD | Normal | 9709 | 2 |
| | Attacks | 3 | 12830 |
| UNSW-NB15 | Normal | 56098 | 2 |
| | Attacks | 5 | 119336 |
| KDD-CUP | Normal | 396740 | 3 |
| | Attacks | 6 | 97273 |
| CSE-CIC-IDS2018 | Normal | 526038 | 1 |
| | Attacks | 6 | 522532 |

same procedure is applied for decryption as well. Subsequently, a Bitwise-XOR operation is performed between the ciphertext and the pseudorandom array generated in the earlier step to obtain the plaintext.

## IV. SIMULATION OUTCOMES AND ANALYSIS

This section explains the simulation outcomes and analysis for the AD and PPKM models implemented in MATLAB. The detection outcomes are calculated based on the positive and negative measures. The reproducibility of the experiments, particularly concerning algorithm initialization, is ensured through detailed documentation and transparent reporting of the methodology. Specific steps in initializing algorithms are clearly described, including parameter settings and initialization conditions. This approach enables other researchers to accurately replicate the experimental setup of the proposed model as illustrated in Table II.

### A. VECGLSTM Comparison

The performance analysis of the introduced VECGLSTM model is illustrated in this section. Table III illustrates the confusion matrix curves generated by the introduced model when applied to the datasets NSL-KDD, UNSW-NB15, KDD-CUP, and CSE-CIC-IDS2018. These curves provide a visual representation of the detection performance, as they showcase the alignment between actual and predicted values as derived from the confusion matrix outputs.

Tables IV–VI, and Fig. 2 highlight VECGLSTM's performance. For NSL-KDD, VECGLSTM achieves 99.8% accuracy, high recall, precision, F1 score, and detection rate, with an MCC of 89%. On UNSW-NB15, it reaches 99.79% accuracy, 100% recall, precision, F1-score, and a 99.01% MCC. For KDD-CUP, VECGLSTM attains 99% accuracy, outperforming HESL, EICDL, and QNN in all metrics.

Fig. 3 shows the dynamic relationship between EGOA's performance and the number of iterations, with the fitness curve generally descending and improving over time. This trend demonstrates EGOA's ability to enhance solutions effectively, providing empirical proof of its slower processing time compared to algorithms like GOA [41], Bat Optimization Algorithm (BOA) [42], and Adaptive Rat Swarm Optimization (ARSO) [43], as shown in Fig. 4.

The performance of the introduced VECGLSTM is evaluated with recent traditional approaches like HIDM [17], G-ABC with DNN [18], EICDL [19], MLP-PSO [20], HESL [21], HEML [22], QNN [23] and Hyb EESCCNN [24]. The proposed model is evaluated against several performance measures like detection rate, recall, precision, F1-score, accuracy, AUC, and Matthews Correlation Coefficient (MCC).

Table VII presents the sensitivity analysis of correlation-based feature selection thresholds and their impact on the attack detection model's performance for datasets NSL-KDD, UNSW-NB15, KDD-CUP, and CSE-CIC-IDS2018. The optimal threshold is around 0.5, where the highest accuracy is achieved for most datasets. At this threshold, the model balances feature relevance and redundancy, effectively selecting informative features and discarding irrelevant ones. Lower thresholds include too much noise, while higher thresholds omit essential features, reducing model performance. Thus, the 0.5 threshold offers the best feature set, enhancing the model's accuracy in detecting attacks across different datasets.

### B. CCPTSO Comparison

Similarly, the performance of the introduced cryptographic framework is compared with recent privacy modules like ATBGVHE [24], ECC [23], and ECC-AES [25].

Table VIII provides a comparative analysis of different cryptographic techniques based on throughput, delay, and training time. The CCPTSO demonstrated improved performance by achieving higher throughput and lower delay compared to existing techniques. It exhibited a throughput of 900 Kbps, indicating better data processing efficiency, while maintaining a delay of 0.29s, which is significantly lower than other methods, making it more suitable for real-time applications. Although its training time of 1.8s is slightly higher than some techniques, it remained within an optimal range for practical implementation.

Fig. 5 provides a comprehensive comparison across different key sizes (ranging from 64 to 256 bytes) for various detector types (ECC, ECC-AES, CAHSACM, ATBGVHE), encryption algorithms, and decryption times. The relationship between key size and detection rate (depicted in Fig. 5(a)) shows a positive correlation, where larger key sizes result in higher detection rates.

However, as illustrated in Fig. 5(b) and 5(c), larger key sizes also lead to increased encryption and decryption times due to the complexity of the mathematical processes involved. This

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TABLE IV
(A) ACCURACY (B) RECALL (C) PRECISION (D) F1-SCORE (E) MCC AND (F) DETECTION RATE COMPARISON USING NSL-KDD

| Performance Measures | HIDM [17] | HESL [21] | G-ABC With DNN [18] | Hyb EESCCNN [24] | VECGLSTM (Proposed) |
|---|---|---|---|---|---|
| **Accuracy (%)** | 99.8 | 97.21 | 98.94 | 99.154 | 99.92 |
| **Recall (%)** | 99.8 | 78 | 99.8 | 100 | 100 |
| **Precision (%)** | 99.8 | 93 | 98.99 | 99 | 99.98 |
| **F1-score (%)** | 99.8 | 96.02 | 99.36 | 99.672 | 99.99 |
| **MCC (%)** | 89 | 91.2 | 98 | 98 | 97.8 |
| **Detection rate (%)** | 99.8 | 96.43 | 97 | 98 | 99.89 |

TABLE V
(A) ACCURACY (B) RECALL (C) PRECISION (D) F1-SCORE (E) MCC AND (F) DETECTION RATE COMPARISON USING UNSW-NB15

| Performance Measures | HIDM [17] | G-ABC with DNN [18] | Hyb EESCCNN [24] | HEML [22] | VECGLSTM (Introduced) |
|---|---|---|---|---|---|
| **Accuracy (%)** | 80.4 | 98.54 | 99.67 | 96.25 | 99.79 |
| **Recall (%)** | 80.4 | 99.34 | 100 | 96.80 | 100 |
| **Precision (%)** | 82.5 | 99.12 | 100 | 98.75 | 100 |
| **F1-score (%)** | 81.3 | 99.107 | 100 | 97.76 | 100 |
| **MCC (%)** | 88.86 | 98 | 99 | 96 | 99.01 |
| **Detection rate (%)** | 92.7 | 97 | 98 | 94 | 98.76 |

TABLE VI
(A) ACCURACY (B) RECALL (C) PRECISION (D) F1-SCORE (E) MCC AND (F) DETECTION RATE COMPARISON USING KDD-CUP

| Performance Measures | HESL [21] | EICDL [19] | QNN [23] | VECGLSTM (Introduced) |
|---|---|---|---|---|
| **Accuracy (%)** | 95.41 | 97.88 | 98.9 | 99 |
| **Recall (%)** | 94 | 93.51 | 97.2 | 98.07 |
| **Precision (%)** | 93.01 | 95.18 | 98 | 98.85 |
| **F1-score (%)** | 95.11 | 94 | 96.7 | 98.47 |
| **MCC (%)** | 89.06 | 97 | 96 | 97.93 |
| **Detection rate (%)** | 95.19 | 96.9 | 97.98 | 96.38 |
| **AUC** | 0.912 | 0.89 | 0.989 | 0.99 |

highlights a trade-off between security and performance, where larger key sizes offer stronger encryption but necessitate extra time for processing. Overall, the figures collectively demonstrate the impact of key size on detection rates, encryption times, and decryption times across different cryptographic algorithms and detector types.

Fig. 5(d) and 5(e) shows that CCPTSO, the introduced scheme, shows a noticeable performance improvement, especially at higher key sizes, with reduced computational cost and energy consumption compared to other schemes, highlighting its efficiency in cryptographic operations. Overall, collectively demonstrates the impact of key size on detection rates, encryption times, decryption times, computational cost, and energy consumption across different cryptographic algorithms and detector types.

❖ *Ablation Experiment:* Table IX compares chaotic cryptography with and without CCPTSO across key sizes (64, 128, 192, 256 bits). At a key size of 128 bits, CCPTSO significantly improves cryptographic performance: encryption time decreases from 187.6 to 102.5 seconds, decryption time from 277.6 to 199.07 seconds, and processing time from 2.993 to 2.032 seconds. These results underscore CCPTSO's benefits in enhancing efficiency and speeding up cryptographic operations in chaotic cryptography across different key sizes.

### *C. Case Study: Real-Time Hardware Implementation for Cloud Security in a Dynamic Cloud Environment*

During peak shopping periods like Black Friday, an online retail company scales cloud resources dynamically while maintaining security. They implement a real-time hardware setup using Xilinx Virtex UltraScale+ Field Programmable Gate Arrays (FPGAs) for the VECGLSTM model and Mellanox ConnectX-6 Dx smart Network Interface Cards (NICs) NICs for the CCPTSO algorithm, ensuring secure and efficient data processing.

Real-time user and transaction data are preprocessed using the DRFLLS method, handling missing values and ensuring consistency. Relevant features are selected with CFS. The VECGLSTM model on the FPGAs performs real-time anomaly detection and adjusts parameters dynamically, eliminating the need for extensive retraining.

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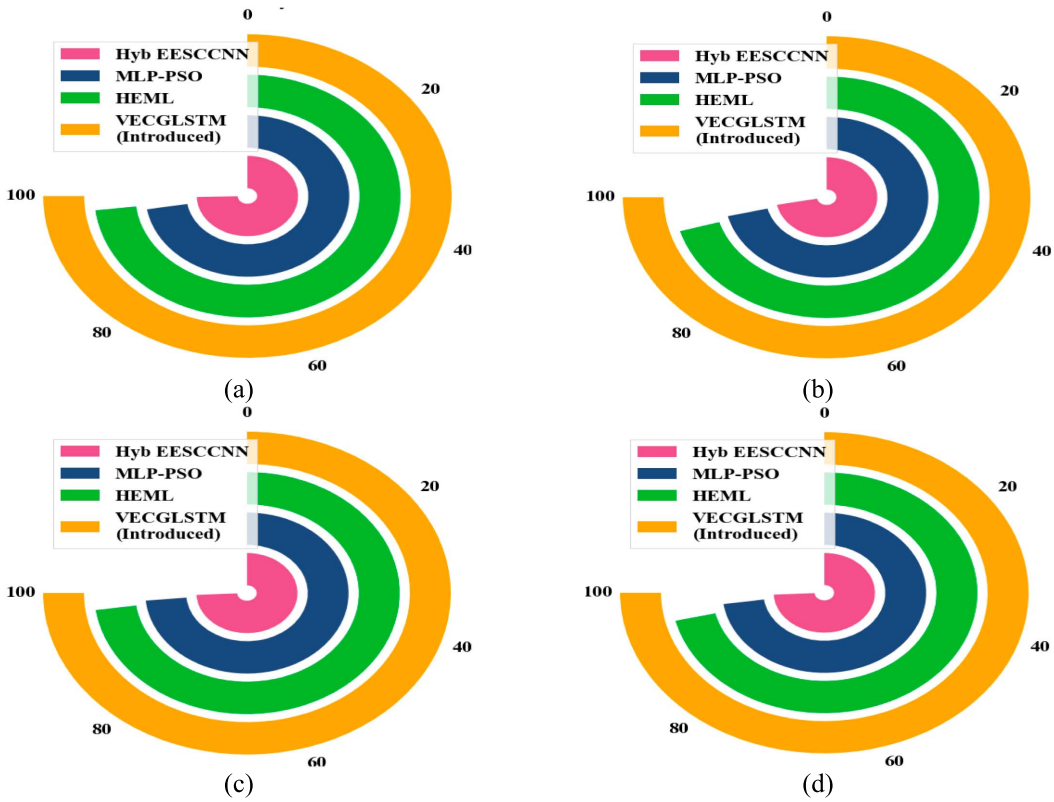



Fig. 2. (a) Accuracy (b) recall (c) precision and (d) F1-score comparison using CSE-CIC-IDS2018.

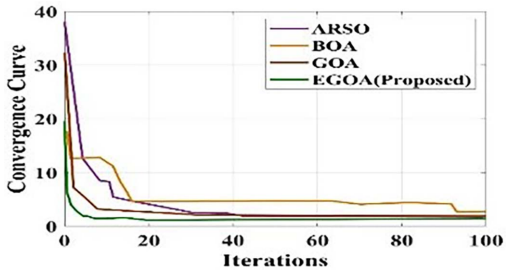



Fig. 3. Fitness evaluation of EGOA.

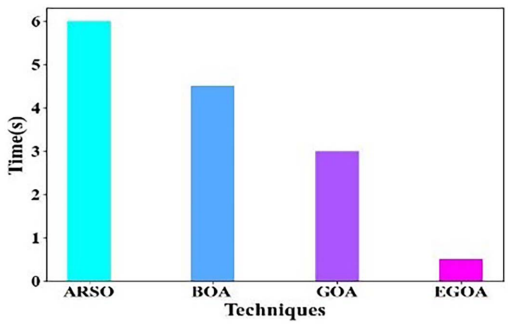



Fig. 4. Time analysis of EGOA.

For the experimental setup, the following values and configurations are used:

- **FPGAs:** Xilinx Virtex UltraScale+ VU9P, configured with a clock speed of 200 MHz, capable of processing up to 1 million transactions per second.

TABLE VII
EFFECT OF CORRELATION-BASED FEATURE SELECTION THRESHOLDS ON ACCURACY

| Threshold | Accuracy (%) | | | |
|---|---|---|---|---|
| | NSL-KDD | UNSW-NB15 | KDD-CUP | CSE-CIC-IDS2018 |
| **0.1** | 94.1 | 92.5 | 93.2 | 91.8 |
| **0.2** | 95.3 | 93.8 | 94.5 | 92,7 |
| **0.3** | 96 | 94.2 | 95 | 93.3 |
| **0.4** | 96.4 | 94.7 | 95.4 | 93.9 |
| **0.5** | 99.92 | 99.79 | 99 | 99.64 |
| **0.6** | 96.5 | 94.8 | 95.5 | 93.8 |
| **0.7** | 96.2 | 94.5 | 95.3 | 93.5 |
| **0.8** | 95.8 | 94.2 | 95 | 93.1 |
| **0.9** | 95.5 | 93.9 | 94.7 | 92.8 |

TABLE VIII
CCPTSO COMPARISON

| Techniques | Throughput (Kbps) | Delay (s) | Execution Time (s) |
|---|---|---|---|
| **ATBGVHE [23]** | 800 | 0.5 | 1.32 |
| **ECC-AES [24]** | 648 | 2.7 | 2.01 |
| **ECC [22]** | 784 | 3.01 | 1.11 |
| **CCPTSO (introduced)** | 900 | 0.29 | 1.8 |

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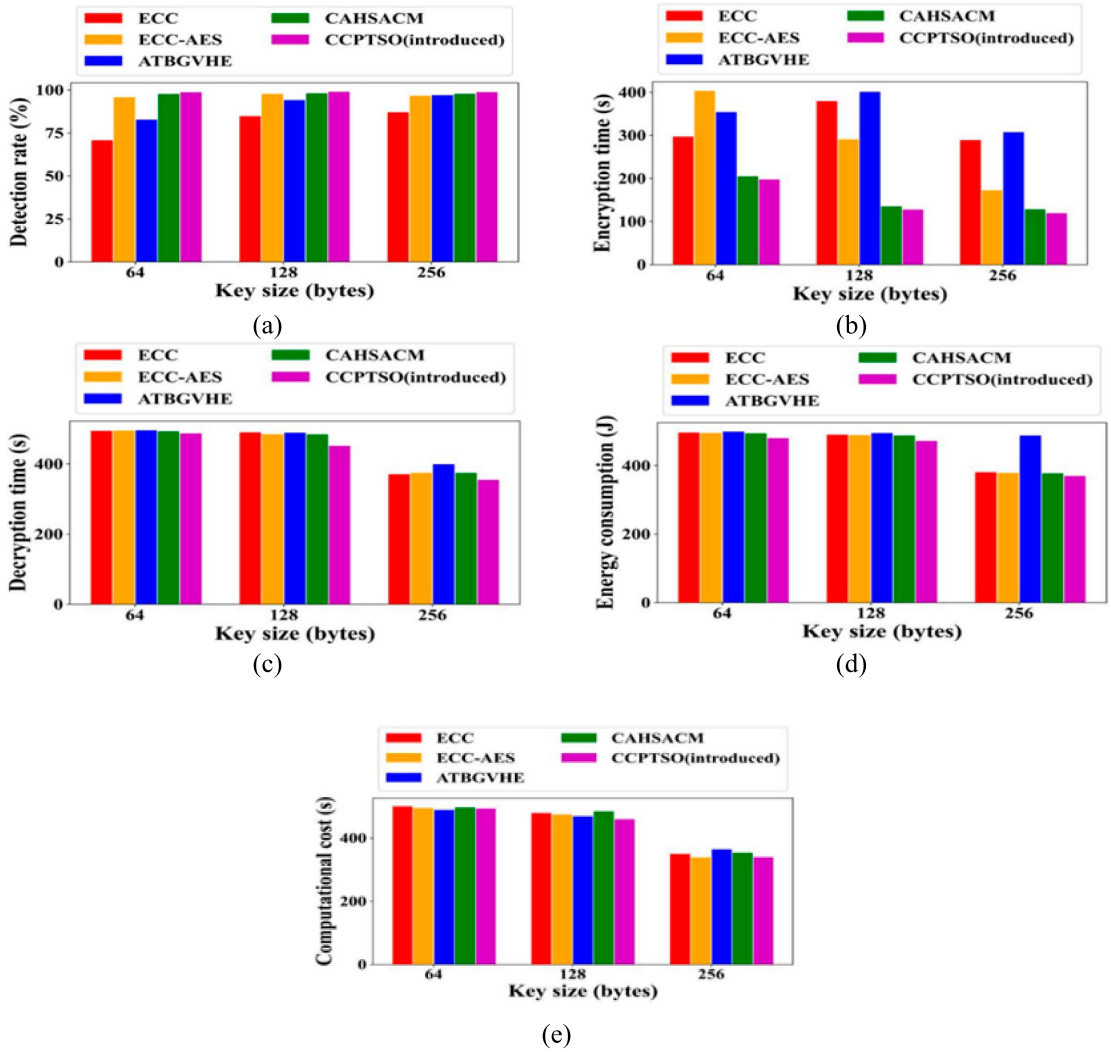



Fig. 5. (a) Detection rate (b) encryption and (c) decryption time (d) energy consumption and (e) computational cost analysis concerning various key sizes.

TABLE IX
ABLATION EXPERIMENT

| | Comparison | Key Size (bits) | | | |
|---|---|---|---|---|---|
| | | 64 | 128 | 192 | 256 |
| **Chaotic Cryptography without CCPTSO** | Encryption time (s) | 143.08 | 187.6 | 190.3 | 193.03 |
| | Decryption time (s) | 264 | 277.6 | 293.9 | 336.4 |
| | processing time (s) | 2.092 | 2.993 | 3.01 | 3.34 |
| **Chaotic Cryptography with CCPTSO** | Encryption time (s) | 99.76 | 102.5 | 111.5 | 150.4 |
| | Decryption time (s) | 189.2 | 199.07 | 274.3 | 289.46 |
| | processing time (s) | 1.276 | 2.032 | 2.558 | 2.614 |

- **SmartNICs:** Mellanox ConnectX-6 Dx, supporting data rates of up to 100 Gbps, ensuring high-speed data transmission and low-latency processing.
- **Data Preprocessing:** The DRFLLS method is set to generate 10 similarity records per missing value, with Pearson correlation and Normalized Root Mean Squared Error (NRMSE) thresholds of 0.9 and 0.05, respectively.
- **Feature Selection:** The CFS method applies a correlation threshold of 0.5 for feature selection.

The system undergoes load testing by simulating peak shopping periods and monitoring performance metrics such as latency, throughput, and threat detection accuracy. Simultaneously, the CCPTSO algorithm on Mellanox ConnectX-6 Dx SmartNICs manages secure data transmission, encrypting and decrypting transaction data to maintain integrity and confidentiality. Security tests confirm the system's effectiveness in mitigating cyber threats during scaling. This case study demonstrates the proposed model's adaptability in a dynamic cloud environment. By

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leveraging FPGAs and NICs for real-time threat detection and secure data handling, the platform scales resources to meet demand without extensive retraining, ensuring robust security and seamless scalability.

## V. Conclusion

This research has successfully developed and tested new deep-learning models for detecting and preventing cyberattacks in cloud environments. The models are based on the VECGLSTM network and a CCPTSO algorithm. The outcomes show that these innovative approaches significantly improve the security of cloud computing environments. The models are evaluated against various types of attacks and outperformed existing methods. The VECGLSTM network proved effective in detecting and mitigating cyberattacks, while the CCPTSO model ensured data confidentiality in cloud systems with high throughput (900 Kbps) and low delay (0.29s). In the future, to enhance performance under adversarial conditions in real-time, the proposed method will integrate adversarial training techniques during model training. This involves exposing the model to dynamically generated adversarial examples to improve resilience against evasion tactics that mimic normal user behavior. Additionally, incorporating advanced anomaly detection algorithms capable of real-time processing, such as streaming anomaly detection methods or online learning approaches, will boost the model's ability to identify subtle deviations from normal patterns instantly. Continuous updates and refinement of training data with diverse adversarial scenarios will ensure the model remains adaptive and effective in detecting evolving threats, thereby bolstering its robustness in real-time cybersecurity applications.

## Data Availability Statements

The data supporting the findings of this study are cited and the link is listed in the references.

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## Conflict of Interest

The authors declare that they have no potential conflict of interest.

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